SPATIAL ANALYSIS OF AN ORCHARD BASED ON UAV RGB IMAGERY AND GIS TECHNOLOGIES/ ANALIZA SPAȚIALĂ A UNEI LIVEZI PE BAZA DATELOR UAV RGB ȘI A TEHNOLOGIILOR GIS

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ABSTRACT

"Precision agriculture increasingly relies on unmanned aerial vehicles (UAVs) and GIS techniques for the high-resolution spatial characterization of vegetation. This study evaluates the use of UAV RGB imagery, combined with GIS techniques, for the individual and spatial characterization of trees in an orchard. A DJI Mini 2 SE drone was used to acquire aerial images over an area of approximately 600 m², containing 15 fruit trees, the images being processed photogrammetrically in WebODM to generate the orthophoto, DSM and DTM. Three methods of canopy height estimation were compared: CHM obtained as the difference between DSM and DTM, the buffer-DSM approach, and a CHM derived from point cloud classification using the Cloth Simulation Filter (CSF) algorithm in CloudCompare. To characterize vegetation variation, the RGB indices GLI and VARI were calculated at crown level and spatially interpolated using the IDW method. Field validation indicated, for the CSF-based method, a detection rate of 87% and an RMSE of 0.40 m. GLI and VARI values showed variations associated with differences in vegetation condition observed in the field. The results highlight the potential of integrating UAV RGB imagery, photogrammetry and GIS for the preliminary spatial characterization of individual orchard trees."

REZUMAT

„ Agricultura de precizie se bazează tot mai mult pe sisteme aeriene fără pilot (UAV) și tehnici GIS pentru caracterizarea spațială a vegetației la rezoluție ridicată. Studiul de față evaluează utilizarea imaginilor UAV RGB, combinate cu tehnici GIS, pentru caracterizarea individuală și spațială a pomilor dintr-o livadă. O dronă DJI Mini 2 SE a fost utilizată pentru achiziția de imagini aeriene asupra unei suprafețe de aproximativ 600 m², care conținea 15 pomi fructiferi, imaginile fiind procesate fotogrammetric în WebODM pentru generarea ortofotoplanului, DSM și DTM. Au fost comparate trei metode de estimare a înălțimii coronamentului: CHM obținut prin diferența DSM–DTM, abordarea buffer-DSM și un CHM derivat din clasificarea norului de puncte utilizând algoritmul Cloth Simulation Filter (CSF) în CloudCompare. Pentru caracterizarea variației vegetației au fost calculați indicii RGB GLI și VARI la nivelul fiecărui coronament și interpolați spațial prin metoda IDW. Validarea pe teren a indicat pentru metoda bazată pe CSF o rată de detecție de 87% și o eroare RMSE de 0,40 m. Valorile GLI și VARI au prezentat variații asociate cu diferențele de stare a vegetației observate în teren. Rezultatele evidențiază potențialul integrării UAV RGB, fotogrammetriei și GIS pentru caracterizarea spațială preliminară a pomilor individuali."

INTRODUCTION

Precision agriculture is a branch of sustainable agriculture that aims to make more efficient use of agricultural resources and reduce environmental impact. Advances in technology and data analysis capabilities have enabled precision agriculture to support decision-making in farming practices (Sishodia et al., 2020). Traditional vegetation monitoring methods generally rely on physical fieldwork, making them time-consuming and costly. In this context, the increasingly widespread use of UAVs has reshaped agricultural practices, as unmanned aerial vehicles have advanced considerably in recent years and are now equipped with advanced imaging sensors capable of capturing aerial imagery at low cost over large areas (Anzar et al., 2025). Individual tree detection in orchard monitoring allows for improved management practices. In addition, analysing each

tree separately provides structural, geometric and positional information specific to each individual case (Dong et al., 2020). When crown boundaries are delineated with greater precision, the resulting geometric data — crown area and diameter — become more reliable, as the risk of confusion with neighbouring vegetation is reduced, supporting more accurate decisions on irrigation, fertilization and other management practices (Ball et al., 2023). Compared with satellite imagery, UAVs offer a clear advantage in terms of spatial detail and the frequency with which the same area can be surveyed, making them a more flexible solution for farmers (Ponce et al., 2022). However, limitations remain regarding battery life and the maximum area that can be covered within a single orchard (Zhang et al., 2021).

Photogrammetric processing of UAV images, using techniques such as Structure from Motion (SfM) and Multi-View Stereo (MVS), makes it possible to derive several key products, including three-dimensional point clouds, digital orthophoto maps (DOM), digital surface models (DSM) and digital terrain models (DTM) (Dong et al., 2020). These products can subsequently be used to obtain useful information about individual trees, such as height, 3D coordinates, crown diameter and crown area. Among the available approaches, the Canopy Height Model (CHM) is commonly used to estimate tree height, typically calculated as the difference between the digital surface model and the digital terrain model (Vacca and Vecchi, 2024). However, comparing UAV-based height extraction methods on a university campus, Hao et al. (2022) reported that heights derived from CHM tend to be lower than field-measured values, while height extraction directly from the point cloud achieved a mean absolute error of 10.28 cm, outperforming the CHM-based approach.

UAV RGB image acquisition makes it possible to obtain spectral information from the visible domain. The spatial evaluation and monitoring of vegetation frequently relies on the calculation of spectral indices, which can provide useful information on vegetation status (Gitelson et al., 2002). Although many studies have focused on multispectral and hyperspectral data for vegetation assessment, RGB imagery also provides valuable information that can be used to characterize vegetation variability. Vegetation indices derived from RGB bands, such as the Green Leaf Index (GLI) and the Visible Atmospherically Resistant Index (VARI), have shown potential for detecting variations in plant vigour and early signs of stress (Banaszek and Szota, 2025). For instance, Nur Anisa et al. (2020) demonstrated that the health of oil palm trees could be assessed using the VARI index derived from UAV imagery, highlighting the considerable potential of RGB-based data for vegetation health monitoring.

The present study aims to evaluate the feasibility of a low-cost UAV RGB workflow, combined with open-source GIS software, for the spatial and structural characterization of a small backyard orchard. Specific objectives were to: compare three methods of tree height estimation in terms of detection rate and accuracy against field measurements; calculate and interpret the GLI and VARI vegetation indices at individual tree level; and generate an interpolated spatial representation of index variability across the orchard using IDW.

MATERIALS AND METHODS

The study was conducted in a private garden located in Blejoi, Prahova County, Romania (Figure 1), covering an area of approximately 600 m², which was used as an experimental area for tree analysis using photogrammetric and GIS-based methods. The study area included 15 fruit trees, for which spatial and structural characteristics were subsequently determined and RGB vegetation indices were calculated.



Fig.1. Location of the study area in Blejoi, Prahova County, Romania. (source: WebODM)

Aerial image acquisition was carried out on 11 July 2026, starting at 11:30. Weather conditions at the time of the flight were favourable, with clear skies, an air temperature of approximately 24 °C and a wind speed of approximately 7 km/h; no significant shading of the study area was present during acquisition.
Data were acquired using a DJI Mini 2 SE drone equipped with an RGB camera. The flight was conducted at an altitude of 30 m, with the camera oriented at nadir (90°) and an image overlap of approximately 70–72%, resulting in a theoretical ground sampling distance (GSD) of approximately 1 cm at the flight altitude used. A total of 40 aerial images were acquired for the study area and subsequently used for photogrammetric reconstruction; the final orthophoto exported from WebODM was generated at a resolution of 5 cm/pixel, corresponding to the software's default orthophoto-resolution setting.
To validate the results obtained through photogrammetric methods, direct field measurements of tree height were subsequently carried out approximately one month after the flight date. Measurements were taken using a tape measure, an instrument associated with an estimated error of approximately ±10 cm. The one-month interval between UAV acquisition and field measurements was considered negligible for result comparison, as the expected growth of mature trees over this period is on the order of 1–3 cm.
Aerial images were processed in WebODM, producing an RGB orthophoto (used for colour analysis and tree digitization), a DSM, a DTM, and a raw photogrammetric point cloud (used independently for ground/surface classification, described below). Based on the orthophoto, 15 trees were identified and manually digitized, each assigned a unique identifier together with species, crown geometry and derived attribute values.
Three methods of tree height estimation were compared. The first used the classical canopy height model, obtained as the difference between the digital surface model and the digital terrain model (Vacca and Vecchi, 2024):

$$CHM = DSM - DTM \tag{1}$$

For each tree crown, the corresponding CHM values were extracted to obtain an estimate of tree height. The second method relied solely on the DSM: the maximum DSM value inside each crown was compared with the mean DSM value in an external buffer ring, used as a local estimate of terrain elevation, with height calculated as the difference between the two. The third method was applied directly to the raw point cloud, independently of WebODM's rasterized outputs: ground and vegetation points were separated using the Cloth Simulation Filter (CSF) algorithm in CloudCompare (Zhang et al., 2016). Ground points were rasterized at 0.1 m resolution using Delaunay triangulation to fill small data gaps, with cells lacking nearby ground points explicitly flagged as NoData. A surface model was generated in the same manner from the full point cloud, and the difference between the two produced a third, independently classified CHM dataset.
Two RGB-derived vegetation indices were calculated at crown level and averaged per tree: the Green Leaf Index (Louhaichi et al., 2001):

$$GLI = \frac{2G - R - B}{2G + R + B} \tag{2}$$

and the Visible Atmospherically Resistant Index (Gitelson et al., 2002):

$$VARI = \frac{G - R}{G + R - B} \tag{3}$$

where R, G and B are the reflectance values of the red, green and blue bands, respectively.
Both indices are sensitive to the proportion of green canopy cover and to biochemical changes in leaf pigmentation associated with vegetative stress. Relationships between the two indices and the structural parameters of the trees (height, crown area) were further examined using the Pearson correlation coefficient.
Finally, thematic maps of tree height, species, GLI and VARI were produced in QGIS. The spatial distribution of GLI and VARI was interpolated using Inverse Distance Weighting (IDW) (Shepard, 1968; Sobjak et al., 2023), based on the 15 point values, with results clipped to the boundary of the study area.

RESULTS

To characterize the spatial distribution of the orchard trees, four thematic maps were produced in QGIS: tree height, species, GLI, and VARI (Figure 2). Together, these maps provide an overview of the structural and RGB-derived vegetation characteristics of the analysed trees and allow the spatial variability among individual trees to be visually assessed.

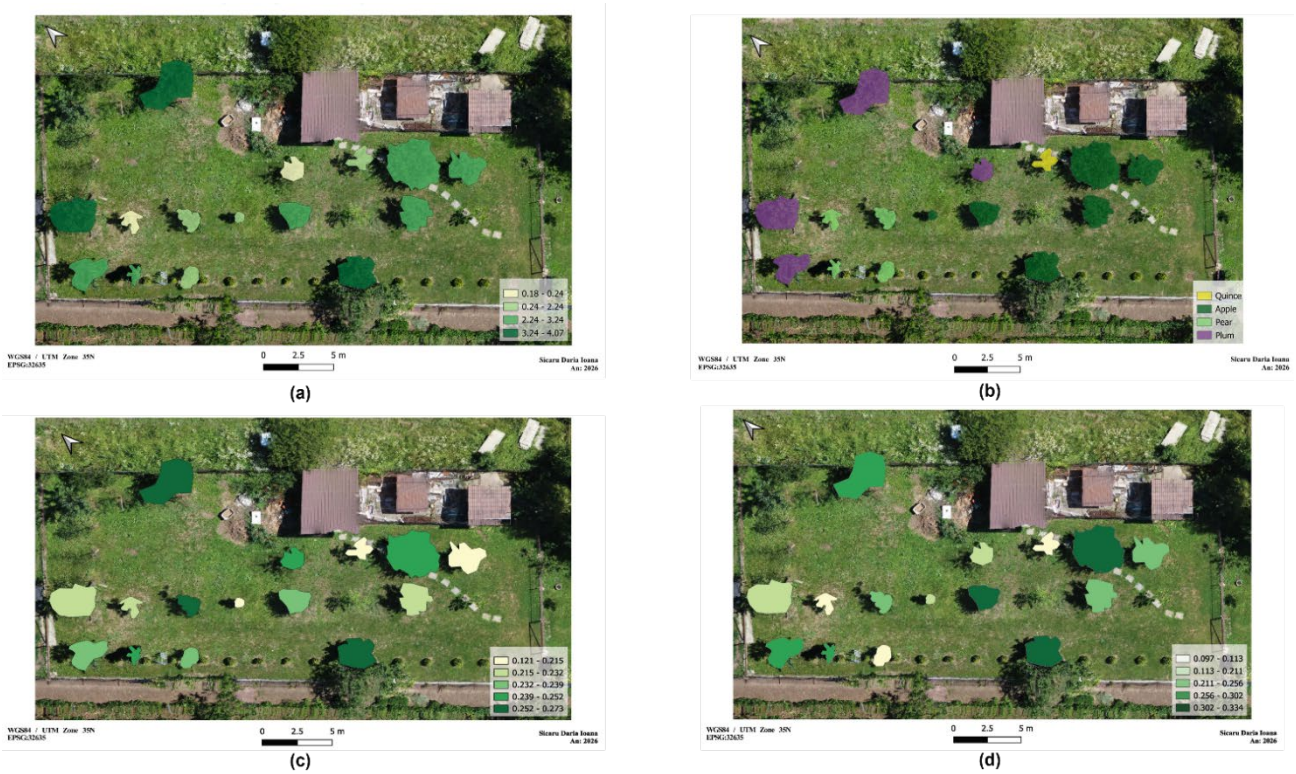


Fig.2. Thematic maps of (a) tree height, (b) species distribution, (c) GLI, (d) VARI

Three methods of tree height estimation were compared, as described in the previous section. Mean heights across the 15 trees were 1.16 m for the CHM method, 2.26 m for the buffer-DSM method, and 2.53 m for the point-cloud-based CHM (CloudCompare/CSF). Detection rate — the proportion of trees for which a method returned a height above a 0.3 m crown-detection threshold — and accuracy indicators (RMSE, MAE, mean bias), computed on the subset of 11 trees detected simultaneously by all three methods, are summarized in Tables 1 and 2.

The CSF-based method achieved both the highest detection rate (87%) and the lowest estimation error (RMSE = 0.40 m, MAE = 0.33 m), likely due to direct control over ground/vegetation point classification and the fine interpolation resolution used (0.1 m), compared with the automatic classification in the WebODM pipeline.

All three methods showed a negative bias, indicating systematic underestimation of tree height, most pronounced for the classical CHM approach; the close match between bias and MAE for each method suggests the error is predominantly systematic rather than random, opening the possibility of a future calibration correction.

Detection failures were concentrated at trees ID 5, 7, 8 and 12, with a progressive improvement across methods: the classical CHM missed 4 of 15 trees, buffer-DSM missed 3, and the point-cloud-based CHM missed only 2 (ID 7 and 8), likely reflecting small crown size, sparse foliage, shading, and the inherent limitations of RGB-only photogrammetric reconstruction. Overall, the CSF-based method was considered the most suitable for height estimation in this dataset.

Table 1

Detection rate per method

Method	Trees detected	Total trees	Detection rate
CHM (DSM − DTM)	11	15	73%
Buffer-DSM	12	15	80%
CSF (CloudCompare)	13	15	87%

Table 2

RMSE, MAE, mean bias per method

Method	RMSE (m)	MAE (m)	Mean bias (m)
CHM	1.85	1.75	−1.75
Buffer-DSM	0.55	0.47	−0.42
CSF (CloudCompare)	0.40	0.33	−0.25

GLI and VARI values were calculated for each of the 15 trees analysed (Table 3), ranging from approximately 0.12 to 0.27 for GLI and from approximately 0.10 to 0.33 for VARI.

Table 3

Mean GLI and VARI values per tree

ID	Species	GLI Mean	VARI Mean
1	Apple	0.21	0.24
2	Apple	0.22	0.26
3	Apple	0.26	0.33
4	Apple	0.24	0.30
5	Quince	0.12	0.11
6	Apple	0.23	0.29
7	Plum	0.24	0.21
8	Apple	0.19	0.20
9	Pear	0.26	0.26
10	Pear	0.24	0.18
11	Pear	0.25	0.27
12	Pear	0.22	0.10
13	Plum	0.24	0.29
14	Plum	0.23	0.23
15	Plum	0.27	0.27

Relationships between the two indices and tree structural parameters were examined using the Pearson correlation coefficient. A moderate positive correlation was found between GLI and tree height ($r = 0.402$), and a stronger positive correlation between VARI and height ($r = 0.672$) (Figure 3). A positive association was also found between GLI and VARI themselves ($r = 0.664$). Given the limited sample size ($N = 15$), these correlation coefficients should be interpreted as indicative trends rather than robust statistical relationships.

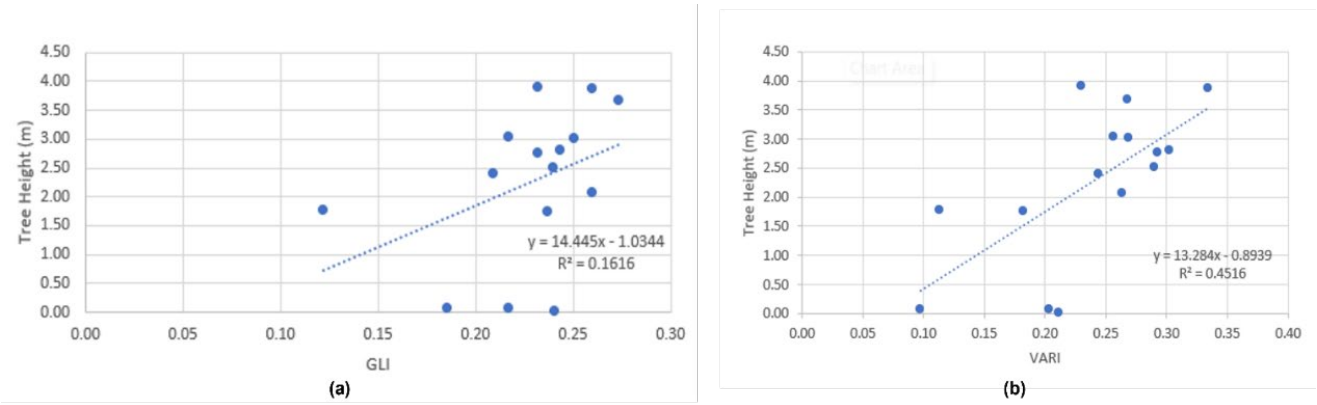


Fig. 3. Relationship between (a) GLI and tree height, and (b) VARI and tree height

Field validation, carried out for the trees showing the lowest index values, confirmed the sensitivity of both indices to canopy health and foliage structure.

Tree ID 5 (quince), which had the lowest values of both indices (GLI = 0.12, VARI = 0.11), showed severe drying, curling and complete browning of the foliage in the field together with very low leaf density (Figure 4a, b). The low GLI reflects the geometric deficit of green canopy mass caused by visible soil between branches, while the low VARI reflects the biochemical degradation of green pigments and increased reflectance in the red band.

Tree ID 12 (pear) showed a contrasting pattern: a normal GLI (0.22), maintained by a well-developed, dense crown that covered the soil, but a markedly reduced VARI (0.10), explained by strong foliar chlorosis, most likely due to nutrient deficiency (Figure 4c, d). Direct comparison with a healthy pear leaf confirmed that VARI responded immediately to this change in leaf hue, illustrating its value as an early indicator of canopy stress not always captured by GLI alone.



Fig. 4. Field validation of spectral anomalies: (a) overall crown view of tree ID 5 (quince); (b) close-up of the foliage of tree ID 5; (c) overall crown view of tree ID 12 (pear); (d) direct comparison between a chlorotic leaf from tree ID 12 (top) and a healthy pear leaf (bottom).

The spatial distribution of GLI and VARI was interpolated using the IDW method described in the previous section, producing continuous surfaces from the 15 point values, clipped to the boundary of the study area (Figure 5). The resulting maps highlight zones where estimated index values are higher or lower depending on the values measured at neighbouring trees, allowing the transition from a point-based to a continuous representation of spatial variability within the orchard.



Fig. 5. IDW-interpolated spatial distribution of (a) GLI and (b) VARI

It should be noted that these maps represent a spatial estimate based on the values available at the analysed points, rather than a direct measurement of the indices at every location within the orchard.

CONCLUSIONS

This study evaluated the potential of a low-cost UAV RGB workflow, combined with open-source GIS tools, for the preliminary structural and spectral characterization of orchard trees. Among the three height-estimation methods tested, the point-cloud-based CHM (CSF algorithm) performed best, achieving the highest detection rate (87%) and the lowest estimation error (RMSE = 0.40 m), while all three methods showed a systematic tendency to underestimate tree height, most pronounced for the classical CHM approach.

The GLI and VARI indices proved sensitive to visible canopy stress observed in the field, with VARI showing a stronger relationship to tree height and reacting more clearly to early chlorosis symptoms, while GLI more directly reflected canopy density and coverage. The spatial interpolation of both indices further illustrated how index variation is distributed across the orchard, beyond the individual tree level.

The main limitations of the study are the small number of trees analysed, the use of a single flight, and the exclusive reliance on RGB imagery rather than multispectral data. Future work should extend this workflow to a larger and more diverse orchard, incorporate multispectral sensors, and automate data processing using Python.

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